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RESEARCH ARTICLE

A Comprehensive Analysis of Classic Machine Learning and Deep Learning Modeling for Breed Recognition of Bananas

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ABSTRACT Bananas are among the world's most widely farmed, traded, and consumed fruits. Bananas contain several vitamins and minerals, including vitamin B6, vitamin C, magnesium, and manganese. Bananas are a very good source of vitamins and minerals in a densely populated and lower-middle-income country like Bangladesh since they are widely cultivated and available in urban and rural markets all over Bangladesh. Deep learning algorithms have shown considerable promise in recent years for image-based breed recognition tasks, encouraging researchers to look into their potential for breed-specific banana recognition. This research thoroughly investigates the use of deep learning and classical machine learning for banana breed recognition. Seven separate deep learning models were used to recognize bananas: VGG16, ResNet50, Mobile-Net, Inception-v3, MobileNetV3, Hybrid CNN-ViT, and a custom convolutional neural network (CNN), and five classical machine learning classification models: k-NN, SVM, Random Forest, Decision Tree, and Ensemble methods (SVM + Random Forest). These models were trained using images of several banana breeds to differentiate between them based on visual characteristics such as size, shape, color, texture, and skin pattern. The performance of each deep learning and classical machine learning classification model was thoroughly evaluated and tested. The customized CNN model has the highest accuracy of all models. We rigorously tested our models by comparing accuracy along with other key metrics such as recall, precision, and F_1-score. In the cases of these metrics, the customized CNN model outperforms all other deep learning and classical machine learning classification models with an accuracy of 99.37%.

INDEX TERMS Banana breed, deep learning, pre-trained model, customized CNN model, performance metrics, accuracy, artificial intelligence, food technology, image databases, prediction algorithms.

I. INTRODUCTION

Bananas are among the most widely consumed fruits globally, holding a crucial place in many nations' diet and economy. Originating in Southeast Asia, people have been farming bananas for thousands of years, spreading their cultivation to tropical and subtropical areas of the world. A total of 130 nations cultivate and grow bananas, with 79 million metric tons produced in 2009. However, a few nations produce most of the world's bananas, with Brazil, Ecuador, China, India, and the Philippines producing more than 60% of all bananas [1]. The crop has a long history, having been domesticated and cultivated to be essential to the growth of ancient cultures. Bananas are now important export commodities and staple foods in many world regions, including Latin America, Asia, and Africa. Bananas are a good source of essential nutrients, including vitamins and minerals. Vitamins and minerals like vitamin B6, vitamin C, magnesium, and manganese can be found in bananas. Bangladesh produced 826,179 tons of bananas in 2022.

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Bangladesh's production of bananas has been somewhat fluctuating recently, although it generally increased from 1973 to 2022, reaching 826,179 tons in 2022 [2]. Taiwan's Banana Research Institute (TBRI) created Tai-Chiao No. 1, a Cavendish banana variety resistant to race 4 of Panama disease, which considerably increased Taiwan's banana exports in the 1990s [3].

Banana breed recognition offers several benefits, particularly in agriculture and food production. It helps recognize specific banana breeds with unique qualities like taste, size, and disease resistance. This recognition aids in improving breeding programs, ensuring consistent crop yields, and promoting sustainability. It also allows farmers to select the best-suited varieties for different climates and soil conditions, optimizing productivity. Furthermore, recognizing banana breeds helps preserve genetic diversity, crucial for maintaining a healthy banana population amidst challenges like climate change and pests. Ultimately, it improves the efficiency of banana production and enhances the quality of bananas available to consumers. Machine learning and deep learning can assist computers in recognizing banana breeds. Computers may learn to recognize visual traits of various breeds, like fruit size and color, using algorithms trained on a vast dataset of banana images. This enables automated and accurate breed recognition. Farmers can make better judgments about planting, harvest, and distribution by utilizing deep learning, which can greatly increase recognition speed and accuracy. Fig. 1 shows banana cultivation and selling.

Deep learning in recent years has opened the door to tasks that, until now, required high accuracy and efficiency, such as breed recognition. As a sub-area of machine learning, deep learning operates at considerable power concerning image recognition in different domains of interest, from medical imaging to autonomous driving and agricultural applications. Agriculture uses deep learning models for various tasks, including disease detection, yield prediction, and crop classification. However, the number of works carried out on applying deep learning for banana breed recognition is very limited, though its potential may prove revolutionary in this industry.

Farmers can upload images of different banana breeds via a mobile app, which will automatically recognize them in the system. By recognizing the banana varieties, they can take preventive measures against diseases according to the banana breed. This process is quicker than traditional methods. This recognition system offers several practical benefits to key agricultural stakeholders:

- For farmers, it enables timely and breed-specific disease prevention and fertilization strategies, ultimately reducing crop loss and increasing productivity. Agritech companies and agricultural extension services can use the data to generate targeted insights, enabling region-specific advisories.
- Exporters and quality control agencies can rely on the system to ensure breed authenticity, uniformity, and traceability, which are critical for meeting international phytosanitary and labeling standards. By ensuring high-quality and correctly categorized exports, the system boosts buyer confidence and helps farmers penetrate premium global markets.
- The system enhances operational efficiency, reduces manual labor, and enables real-time traceability from farm to market. It also supports data-driven decision-making for banana breeding programs and supply chain management.

(a) Banana cultivation.

(b) Banana selling shop.

FIGURE 1. Banana cultivation and selling (a) Banana cultivation and (b) Banana selling shop.

This will promote the cultivation of high-quality, high-yielding bananas that can withstand harsh weather conditions. Automatic banana breed recognition helps crop management by tracking different breeds of banana and supports market targeting by sorting bananas for the right buyers. Farmers can meet the country's demand and export their bananas abroad, which will help increase the nation's Gross Domestic Product (GDP).

The challenge we're working on is image recognition using deep learning. Deep learning algorithms are reliable for distinguishing between distinct crops and recognizing agricultural diseases. Banana breed recognition is a uniquely challenging problem due to high intra-class

similarity between breeds, subtle visual differences in size, curvature, color, and texture, and environmental variations that affect appearance. Addressing these challenges is vital for ensuring accurate identification, promoting genetic diversity, and improving productivity in banana cultivation. As a result, we may use powerful deep-learning algorithms to differentiate between banana breeds. In order to insightfully view this problem and its details, its flow has been done; the domain is yes so far. This work introduces one of the first curated and annotated benchmark datasets for banana breed recognition, enabling reproducibility in this underexplored domain. We also perform a comprehensive comparison of seven deep learning models. The seven models are ResNet50, VGG16, Mobile-Net, Inception-v3, MobileNetV3, Hybrid CNN-ViT, and a customized CNN model. Analytical accuracy was superior for the unusual CNN model compared to the others. Although Inception-v3 is very close to a customized CNN model, ResNet50 has the lowest level of recognition accuracy for banana breeds. In essence, this research effort's main contributions are as follows:

- To begin addressing the problems of automated banana breed recognition by utilizing deep learning models.
- To collect reliable image data for categorizing banana breeds using alternate CNN/classification algorithms.
- To enable quicker application of disease-resistant treatments, enhancing high-quality, long-lasting, and high-yield banana cultivation
- To conduct an extensive in-depth study on CNN models for banana breed recognition in order to guide future research.
- To curate and annotate a unique image dataset of banana breed, establishing a benchmark for future machine learning and deep learning research in phenotyping.

II. RELATED WORK

This section highlights some of the most recent and significant developments in deep learning. Its influence on agriculture encompasses crop monitoring and recognition. Additionally, it examines intriguing advancements in artificial intelligence.

Shuprajhaa et al. [4] presented a deep learning method for classifying banana ripeness into four stages: unripe, underripe, ripe, and overripe. The technique combines a CNN with eXtreme Gradient Boosting (XgBoost) and linear discriminant analysis to reduce reliance on large datasets. This CNN-XgBoost model achieves an accuracy of 91.25%, outperforming traditional classifiers such as Support Vector Machines, Gaussian Naive Bayes, and k-Nearest Neighbors. Le et al. [5] proposed a novel deep-learning model for clustered banana classification using Mask R-CNN, with an overall accuracy of 96.5%. The model outperforms earlier techniques by identifying the normal and reject classes from a single picture feature more accurately and precisely. Baglat et al. [6] systematically reviewed 1,548 papers on banana ripeness, finding that using banana fingers, sensor cameras, and color features enhances detection accuracy. Their research on four distinct ripeness phases resulted in higher accuracy and coefficient values. Convolutional neural networks excelled with large datasets, whereas standard neural networks and SVMs performed better with sensor data. The assessment also revealed significant gaps in data and methodology within existing studies. Neupane et al. [7] proposed a deep learning method utilizing UAV-captured, high-resolution RGB pictures to count and identify banana plants. Plant detection was accomplished with up to 99% recall using the approach, which combined findings from several altitudes and enhanced photos. Ye et al. [8] show that UAV-based multispectral imaging may detect Fusarium wilt in banana crops. With vegetation indices, especially those featuring a red-edge band, the research attained an accuracy rate of over 80%. Crop management and disease monitoring benefited from increased detection accuracy from higher spatial resolutions (over 2 m). Vijayalakshmi and Peter [9] suggested a five-layer CNN model for banana recognition that achieves 96.98% accuracy when paired with random forest and k-NN classifiers. The model performs better at detecting fruits such as apples, strawberries, oranges, and mangoes than current HOG-based techniques.

Khan et al. [10] demonstrated a method based on deep learning for detecting apple diseases using a dataset of 9,000 annotated RGB photos. The system, enhanced by data augmentation, achieves 88% classification accuracy, making it a valuable tool for early disease diagnosis and control in apples. Future extensions could apply to different crops. Palacios et al. [11] developed a new system that uses computer vision and machine learning to predict early yield in six grapevine varieties. Using SegNet and SVR models, the system showed excellent accuracy in berry detection and yield prediction, with promising cross-validation outcomes. Rahman et al. [12] explored the use of models such as Inception-v3, MobileNet, ResNet50, VGG16, and a customized CNN in deep learning for potato breed detection. The customized CNN achieved the maximum accuracy of 94.84% and outperformed the other models in terms of accuracy, precision, recall, and F_1-score. Khan et al. [13] suggested a CNN-based technique to identify eight distinct hyacinth bean breeds in Bangladesh, emphasizing the high protein and vitamin content of the varieties. Outperforming other models, the customized CNN model had a maximum accuracy of 97.50%. Saha et al. [14] utilized YOLOv8 and EfficientNetB7 models to propose a deep learning-based approach for the early detection of two jackfruit leaf diseases: algal spot and black spot. After training on 6,360 images, the system achieved 99.9% training and 99.5% testing accuracy. By addressing disease-related yield loss, this study demonstrates the potential of AI in increasing agricultural output and boosting global food security. Huang et al. [15] improved the selection process for specialty coffees by creating an automated coffee bean-picking system. The system used a convolutional neural network, image processing, and data augmentation to achieve a 93% recognition rate with a 0.1007 false-positive rate.

Taner et al. [16] used both conventional ML models and CNN-based transfer learning to classify apple types. DenseNet201 achieved an accuracy of 97.48%; when its deep features were merged with SVM, the accuracy increased to 98.28%. Following PCA, MLP increased to 99.77%. The findings support the efficacy of this strategy, and future research will focus on increasing the dataset size and enhancing its diversity coverage. Din et al. [17] developed "RiceNet," a DCNN-based system that outperformed conventional machine learning and pre-trained models in classifying five different types of rice, achieving an accuracy of 94%. It demonstrates how well deep learning works in preventing grain adulteration and ensuring precise rice variety identification. Li et al. [18] traced the origin of jujubes from four different locations of China using neural networks and near-infrared spectroscopy. The two algorithms with the highest accuracy were CNN and LSTM; CNN performed best on smaller datasets and was helpful for non-destructive jujube traceability. Dong et al. [19] developed a channel attention mechanism (MCNN)-based model for mango variety identification using near-infrared reflectance spectroscopy (NIRS), achieving 98.67% accuracy with a quick, non-destructive examination. Escamilla et al. [20] utilized YOLOv5 and DeepSORT to recognize, categorize, and count sweet peppers in greenhouses, achieving a confidence of 0.973 and an accuracy of 85.7% despite occlusions, lighting fluctuations, and variations in maturity level. Kunduracioglu and Pacal [21] fine-tuned CNN and Vision Transformer models for grape disease diagnosis and leaf classification, obtaining 100% accuracy using Swinv2-Base on two grape leaf datasets.

Selvaraj et al. [22] utilized machine learning to classify banana plants and detect diseases in complex African habitats using high-resolution satellite photos and UAV data. The research's mixed-model system and pixel-based random forest model yielded up to 99.4% accuracy, offering a robust decision support system for controlling banana disease in Africa. Fu et al. [23] demonstrated a 99.29% accuracy rate deep learning technique based on YOLOv4 for identifying banana fruits in orchards. YOLOv4 detected different banana sizes and maturities, outperforming YOLOv3 and conventional machine learning in terms of accuracy and speed. Fu et al. [24] described a 100% accuracy and recall RGB camera banana detecting system. The method, which demonstrates excellent accuracy and efficiency, combines background removal, SVM with feature fusion, and multi-scale detection. Sanaeifar et al. [25] developed a computer vision system to track color changes in bananas using RGB and HSV color models, correlating these changes with quality indicators such as pH and hardness. The system utilized an RBF kernel in a support vector regression (SVR) model, achieving greater predictive accuracy than traditional artificial neural networks.

Gong et al. [26] addressed the constraints of classic convolutional neural networks by developing a novel immune convolutional neural network that uses an immune mechanism and GPU acceleration to improve recognition accuracy and processing performance. Mahendran and Vedaldi [27] examine alternative image representations using visualization methods such as inversion, activation maximization, and exaggeration, showing that CNNs maintain detailed, accurate image information through a robust energy-minimalization framework. Hijazi et al. [28] provided a comprehensive overview of CNN and its uses in image identification, highlighting the technology's advantages, high performance, and wide range of applications across several sectors.

Based on the above-mentioned studies, it is visible that no comprehensive investigation into the recognition of banana breeds has been conducted. Furthermore, the majority of studies on fruit and vegetable recognition or leaf disease are subpar and fall short of expectations. So, there is a glaring research gap between the state of research and the current state of the field of study.

III. SYSTEM ARCHITECTURE

A gardener or farmer can use the pre-installed mobile application of the agro-medical expert system on a smartphone or other portable device to take an image of a banana. After that, they enter the picture into the installed program. The system uses its knowledge to make its choice. A server at a distance will store the complete agro-medical expert system. The expert system's installed application can update itself by sending an update request to the server via the Internet. The overall agro-medical expert system is built up of a few interconnected, interrelated components. They are an explanation facility, a knowledge base, an inference engine, and a facility for acquiring knowledge. The inference engine manages the whole agro-medical expert system process. All of the components collaborate to provide the expertise. This expert system component is responsible for producing predictions, suggestions, and answers that are on par with those of a human expert. This system is easy to use, user-friendly, has low computational cost, and is highly deployable. The system can be installed in both market and village environments since it operates on common mobile phones (with Internet) and will give immediate results while in the field; as a result of its compact design, it is very easy to install and to maintain updates. The structural design of the suggested agro-medical expert system for banana breed recognition is displayed in Fig. 2

A. PROBLEM FORMULATION

Given a dataset of N images, each represented as a feature vector:

$$x_i = (x_{i1}, x_{i2}, \ldots x_{in}) \quad 1 \leq i \leq N \tag{1}$$

and corresponding true labels y_i, our deep learning model predicts outputs $f(x_i; \theta)$, where θ represents the parameters of the deep learning model for banana.

The optimization goal is to minimize the error between predicted and true labels:

$$min_\theta \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\left(f\left(x_i; \theta\right), y_i\right) N \tag{2}$$

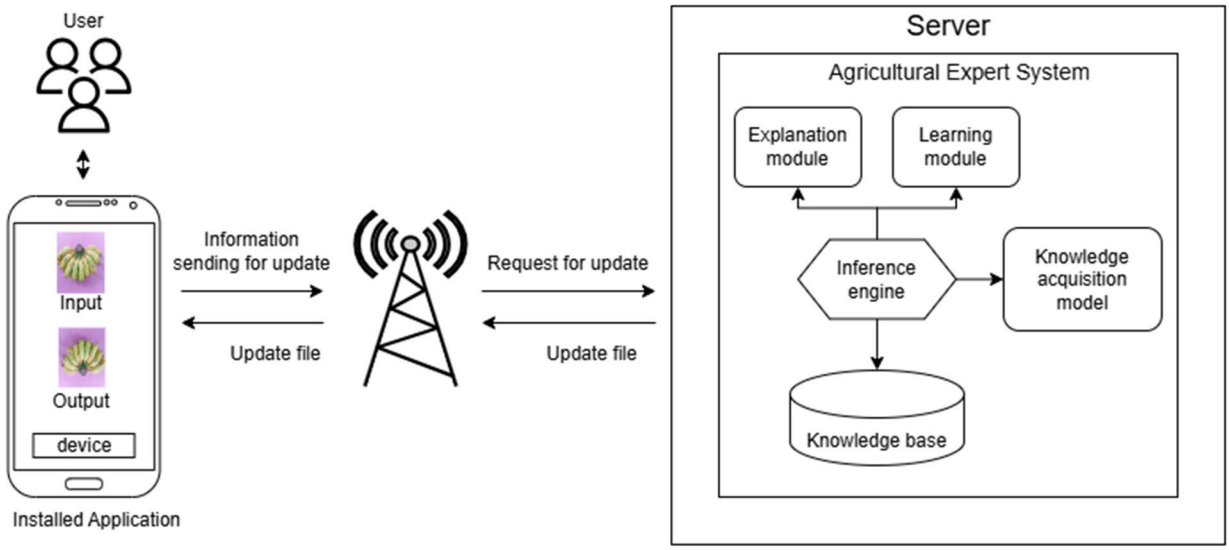


FIGURE 2. The architecture of the banana breed recognition system.

Here:

- $\mathcal{L}\left(f\left(x_i;\theta\right), y_i\right)$ is the chosen loss function (e.g., Mean Squared Error for regression, Cross-Entropy loss for classification).
- $f\left(x_i;\theta\right)$ denotes the predicted output from the deep learning model based on input vector x_i and model parameters θ.

IV. RESEARCH METHODOLOGY

The initial stage of our study technique is data collection. We visited local fruit shops and banana production areas to take images of Bangladeshi bananas. We had to adjust the image sizes because they weren't all the same when we took them. We increased the data set's total number of banana images to 5100, which improved the model's performance. The data collection is prepared for modeling through our cleaning and analysis. This system is capable of handling .jpg files. The holdout approach [29] was applied for the model evaluation, i.e, training, validation, and testing. According to this approach, data is split into three ratios: train, test, and validation. The train is subdivided into another group validation. Then, we implemented our carefully thought-out strategy. After we provide our test data, the model is then evaluated. Fig. 3 describes each point in the block diagram in detail.

A. DATA SET AND DATA PREPROCESSING

A huge number of banana images were required for our testing. We have categorized the demographic information and the regional information. We have collected image data because there is no standard, i.e., publicly available benchmark dataset. The image data was collected on different types of banana breeds. We took images of various typical banana breeds in Bangladesh, visited some neighboring fruit shops, and then headed to the banana field. We also collected some data from the agricultural office. We employed six different banana breeds for our experiment. Table 1 shows an example image of every type of banana.

The first and most important stage is image preprocessing before training the model. Every image was carefully gathered from the field. There are 1268 genuine images in all of our work. Table 2 displays the frequency of images by class. The collected image sizes ranged from 400 to 700 pixels. We collected banana images with a variety of background colors on art paper (blue, green, gray), in an effort to create a consistent background. Our dataset excludes occluded, damaged, or immature bananas. After scaling, each image has a dimension of 224 × 224 pixels.

There is variation in the size of the images we take with our devices. At first, we used image scaling to reduce the data size and solve this issue. After scaling, each image has a dimension of 224 × 224 pixels. After that, traditional deep learning techniques were used to segment the 224 × 224-pixel pictures to extract features. Data augmentation typically involves the addition of various instances to a data collection in increasing numbers to prevent overfitting and improve the deep learning model's resilience. We implemented comprehensive data augmentation techniques to improve the representation of underrepresented breeds and reduce bias. This included transformations with the following parameters: rotation range of 40 degrees, zoom range of 0.2, width shift range of 0.2, height shift range of 0.2, shear range of 0.2, and the ability to flip images horizontally. The 'nearest' fill mode was applied to preserve image borders during augmentation. As a result, we now have 5,100 images to work with. Table 3 provides a visualization of the results of the picture augmentation. Table 2 shows the statistical details of the banana dataset.

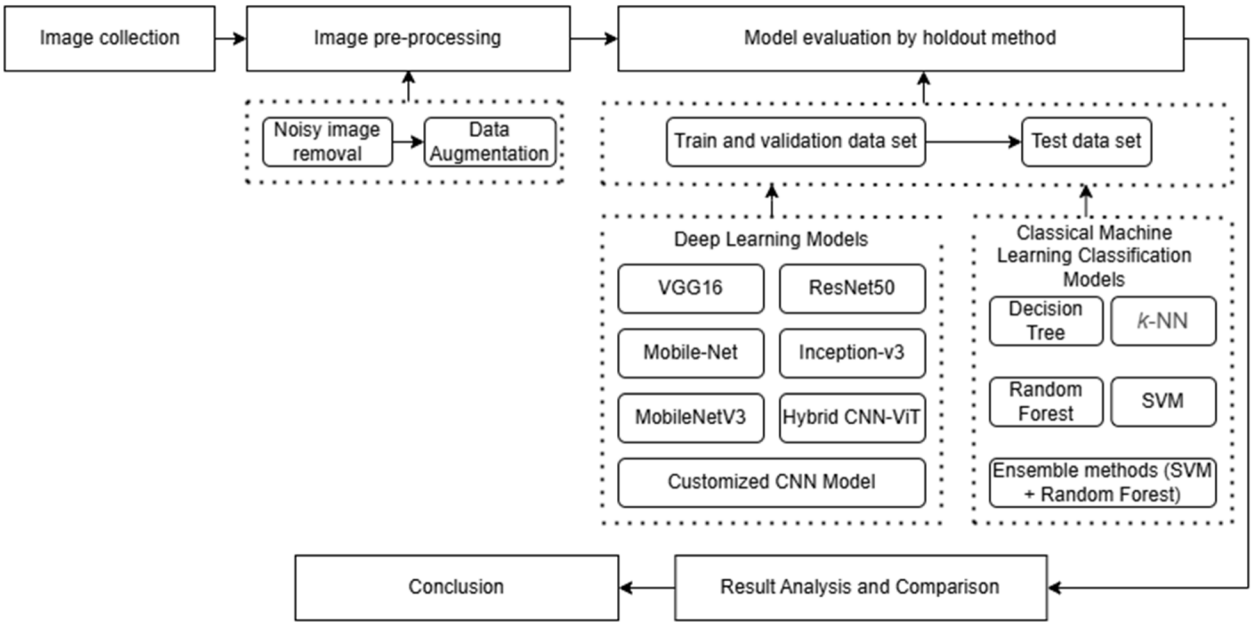


FIGURE 3. Diagram of the working process to recognize banana breeds.

Fig. 4 shows the visualization of statistical details of the banana dataset.

The dataset was split using the holdout approach [29] for training, validation, and testing. Additionally, to ensure model robustness, we also applied k-fold cross-validation and compared results to identify the best-performing model.

B. CLASSIC MACHINE LEARNING CLASSIFICATION MODEL USED

k-Nearest Neighbors (k-NN): The k-NN algorithm is a non-parametric, supervised learning technique that classifies data points according to their closeness in the feature space [30]. For this project, the optimal value of $k = 17$ was selected using a grid search cross-validation approach. A range of odd values of k from 1 to 25 was tested, and $k = 17$ achieved the highest average validation accuracy, making it the most effective choice for our dataset. The distances between points are calculated using the Euclidean distance formula (3):

$$d(D_1, D_2,) = \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2} \tag{3}$$

The essential steps involve loading datasets, extracting features from images, and utilizing Principal Component Analysis (PCA) to reduce dimensionality. Before applying PCA, feature standardization was performed using the standard scaler to ensure that all features contributed equally. PCA was then applied, and we retained 50 principal components that collectively explained approximately 95% of the variance in the data. This balance ensured effective dimensionality reduction without significant loss of information. To ensure consistency, features are standardized with a Standard Scaler, and a grid search is conducted to identify the optimal value of K. After training the best model on the scaled features, predictions are assessed using precision, recall, score, and a confusion matrix, which is presented as a heatmap. Finally,

TABLE 1. An example of each banana breed from the collected dataset.

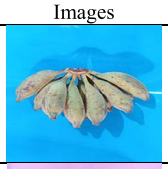
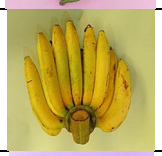
Banana Breeds	Images
Bichi (বিচি কলা in Bengali, pronounced as 'Bichi kala')	
Chini Champa (চিনি চম্পা কলা in Bengali, pronounced as 'Chini Champa kala')	
Kabri (কাবরি কলা in Bengali, pronounced as 'Kabri kala')	
Kacha (কাঁচা কলা in Bengali, pronounced as 'Kacha kala')	
Sagor (সাগর কলা in Bengali, pronounced as 'Sagor kala')	
Subri (সবরি কলা in Bengali, pronounced as 'Subri kala')	

TABLE 2. Statistical details of the banana data set.

Variety Name	Captured Images	Images after Augmentation
Bichi Banana	163	850
Chini Champa Banana	207	850
Kabri Banana	248	850
Kacha Banana	212	850
Sagor Banana	211	850
Subri Banana	227	850
Total	1,268	5,100

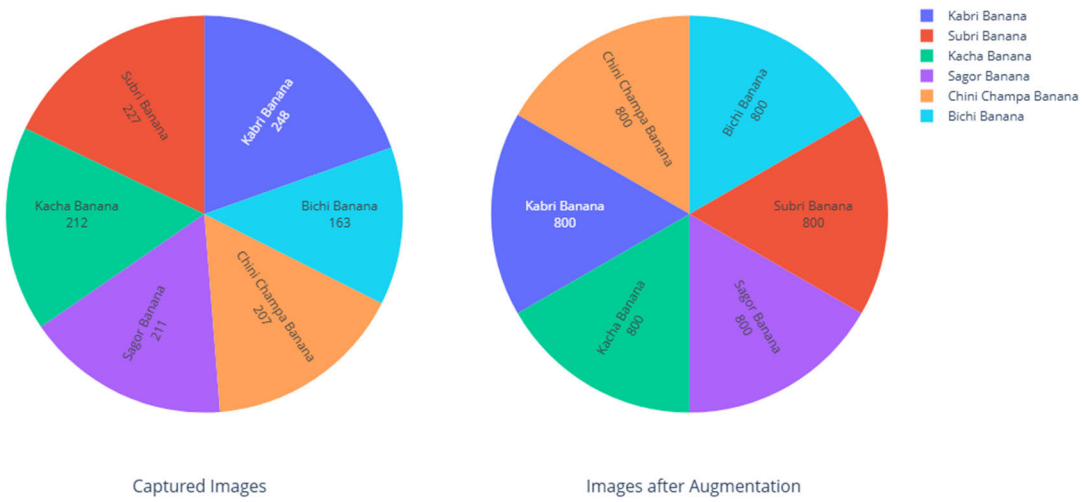


FIGURE 4. Visualization of statistical details of the banana data set.

the trained k-NN model, along with the PCA transformation and scaler, is saved for use in real-time classification tasks. L-time classification tasks [30].

Support Vector Machine (SVM): The SVM model analyzes image data by resizing both training and validation images and extracting relevant features. Through principal component analysis (PCA), these features are condensed to 20 principal components to improve computational efficiency while maintaining key information. A Standard Scaler standardizes the features to ensure uniformity. Hyperparameter tuning is performed using GridSearchCV, which optimizes parameters such as regularization (C), kernel type (linear and RBF), and the kernel coefficient (γ) for RBF kernels [30]. The following equation expresses the decision boundary (4).

$$f(x) = sign(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b) \tag{4}$$

In the SVM model, $K(x_i, x)$ represents the kernel function, α_i are Lagrange multipliers, y_i are class labels, and b is the bias term, defining how data points are classified. Model performance is evaluated using precision, recall, F_1-score and a confusion matrix for visualization. The trained SVM model and PCA transformations and scaling parameters, are saved using joblib for future real-time classification tasks [30].

Random Forest: Random Forest is an ensemble learning technique that builds several decision trees by utilizing random subsets of data and features. This approach minimizes the correlation among the trees, which boosts the model's robustness and generalization abilities [31]. The ultimate prediction is made by a majority vote among all the trees, as illustrated in the equation (5):

$$f(x) = model(f_1(x), f_2(x), \ldots\ldots, f_T(x) \tag{5}$$

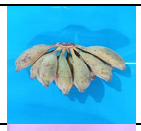
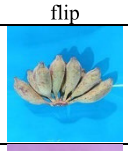
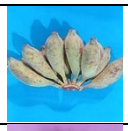
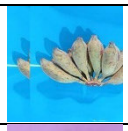
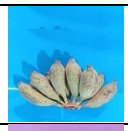
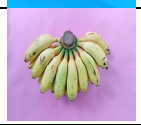
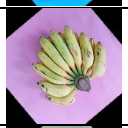
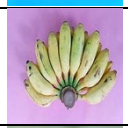
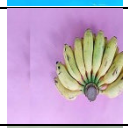
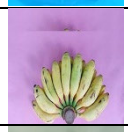
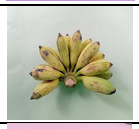
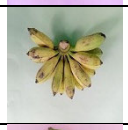
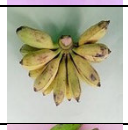
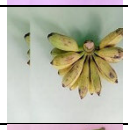
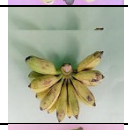
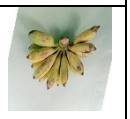
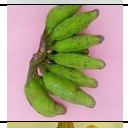
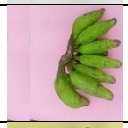
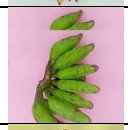
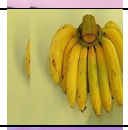
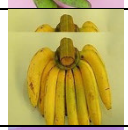
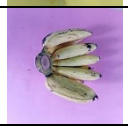
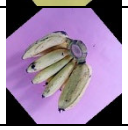
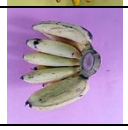
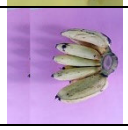
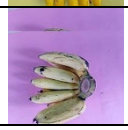
Here, $f_T(x)$ represents the prediction of the T_{th} tree. Random Forest evaluates the importance of features, aiding in the identification of significant predictors within high-dimensional datasets. Its resilience against noise or incomplete data stems from random feature selection and bootstrapping techniques. Additionally, hyperparameter tuning- such as modifying the number of trees and their depth- guarantees optimal performance when discerning complex patterns in fruit disease classification [31].

Decision Tree: The Decision Tree algorithm is a vital non-parametric model that predicts the target variable by recursively splitting datasets according to input features. These splits aim to maximize Information Gain, resulting in the most homogeneous nodes for accurate classifications [32]. The Entropy at node N is calculated as (6):

$$H(N) = -\sum_{i=0}^{k} p_i log_2(p_i) \tag{6}$$

where p_i is the proportion of samples in class i, and k is the total number of classes. The Information Gain from a

TABLE 3. Augmentation effects on each image class.

Banana Breeds	Original Image	Augmented Images					
		Horizontal flip	Rotation: 40 degrees	Zoom: 0.2	Width shift: 0.2	Height shift: 0.2	Shear: 0.2
Bichi							
Chini Champa							
Kabri							
Kacha							
Sagor							
Subri							

potential split S at node N is given by (7):

$$IG(N, S) = H(N) - \sum_{j \in S} \frac{N_j}{N} H(N_j) \tag{7}$$

In practice, during model training, entropy and information gain were calculated at each node to evaluate all possible splits for every feature. The split that maximized the reduction in entropy was selected. To control overfitting and tree complexity, a maximum depth of 10 and a minimum sample split threshold of 5 were applied. These hyperparameters were chosen based on grid search tuning. Furthermore, feature importance was computed as part of the tree-building process. This was achieved by summing the information gain (entropy reduction) contributed by each feature over all splits in which it appeared. The top contributing features included color variance, texture descriptors, and shape ratios, which were most influential in fruit disease classification. Decision Trees are useful baseline models for classifying fruit diseases and offer transparent insights into how decisions are made [32].

Ensemble methods (SVM + Random Forest): The Support Vector Machine (SVM) and Random Forest (RF) ensemble model combines the advantages of both classifiers to improve prediction accuracy. After preprocessing, SVM and RF process the data independently, and the final classification is produced by fusing the outputs of both algorithms using a voting technique. The generalization performance is greatly enhanced by this ensemble method [33]. The voting technique for ensemble modeling is as follows in (8).

$$\hat{y} = mode(\hat{y}_{SVM}, \hat{y}_{RF}) \tag{8}$$

where $\hat{y}_{SVM}$ and $\hat{y}_{RF}$ are the predicted classes by each classifier.

C. CNN MODEL USED

CNNs are a specific type of neural network used in deep learning. The technology of image recognition has advanced significantly with CNN. They are widely used for image evaluation and have been employed in the background of image classification. A CNN consists of fully connected layers, pooling, convolutional, and Rectified Linear Units (ReLU). CNN, on the other hand, uses a well-known architecture. The CNN model's general structure is seen in Fig. 5 [34].

This experiment included seven CNNs: VGG16, ResNet50, Mobile-Net, Inception-v3, MobileNetV3, Hybrid CNN-ViT, and a customized CNN model. VGG16 was created to speed up training while lowering the number of parameters in convolutional layers. There are 138 million parameters in VGG16 [35]. This layer implements many kernels of different sizes. The inception modules included in the Inception network design are the 11th, 55th, 33rd, and max pooling convolutional layers in the network architecture [35]. One of the first organizations to use batch normalization was ResNet50. Convolutional layers and identity blocks are used to construct ResNet50 [35]. The Mobile-Net model filters images using

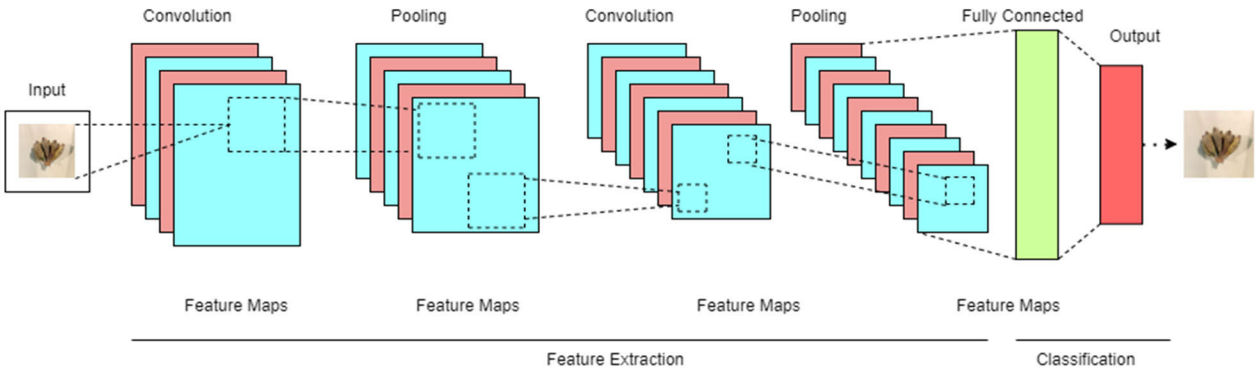


FIGURE 5. General structure of the CNN model for banana breed recognition.

the same convolution as CNN, but does it differently than the original CNN [35]. The bottleneck structure, squeeze-and-excitation module, and depth-wise separable convolution in MobileNetV3 combine to create an intrusion detection model that is both accurate and efficient, while maintaining a lightweight design [36]. Performance in image classification tasks is enhanced by the Hybrid CNN-ViT model, which effectively combines the global context modeling capabilities of Vision Transformers with the local feature extraction capabilities of CNNs [37].

The model's initial layer is a Conv2D layer with 32 filters and a 3×3 size. The ReLU activation function provides the model's non-linearity, which is specified via the activation option. After the initial convolutional layer, there is a second MaxPooling2D layer with a 2×2 pool size. By selecting the greatest value inside every 2×2 pooling zone, this layer reduces the feature map's spatial dimensions. The number of filters steadily increases with regular additions of Conv2D and MaxPooling2D layers. As a result, the model can better extract more complex and ethereal information from the original images. The last MaxPooling2D and Conv2D layers are followed by the flattened layer, which transforms the 2D feature maps into 1D vectors. This is where convolutional layers must be linked to the subsequent fully connected layers. Next, 256 more thick layers with a ReLU activation function are added. This fully connected layer performs high-level feature extraction while adding more non-linearity based on the obtained features from the convolutional layers. Fig. 6 shows the architecture of the customized CNN model.

Deep learning models most commonly use the ReLU activation function. This function returns 0 for negative inputs, but it also returns 0 for all positive x values. As a result, it can be expressed as follows:

$$f(x) = max(0, x) \tag{9}$$

We conducted several experiments to determine the optimal configuration. We systematically adjusted the parameters through hyperparameter tuning, which led us to identify the best configuration for our model.

D. PERFORMANCE METRICS CONSIDERED

Our problem is complex, and a test dataset is used to evaluate performance. Accuracy may not be a reliable indicator of a classifier's actual performance when assessing the performance of classification models constructed from unbalanced datasets with notably different numbers of observations in different classes. As a result, as discussed in [29] and [38], several additional measures are needed to assess a classifier's efficiency and accuracy. In a binary or two-class breakdown, the confusion matrix displays the number of false positives (FPs), false negatives (FNs), true positives (TPs), and true negatives (TNs). However, a multi-class difficulty or issue with more than two classes will produce a confusion matrix that is $n \times n$ ($n > 2$). This matrix, having n columns and n rows, will have an overall count of $n \times n$ items. This matrix cannot calculate the total number of FPs, FNs, TPs, and TNs. We calculated the values of FPs, FNs, TPs, and TNs using the approach provided in [38]. The final confusion matrix is composed of n confusion matrices for the average values of each class, and it has a 2×2 dimension. The confusion matrix calculates the recognition system's accuracy, precision, recall, and F_1-score. A test data set is utilized to assess performance, and these components are used for training. The recall, accuracy, precision, and F_1-score are all calculated using the confusion matrix.

$$\text{Precision} = \frac{TP}{TP + FP} \times 100\% \tag{10}$$

$$\text{Recall} = \frac{TP}{TP + FN} \times 100\% \tag{11}$$

$$F_1\text{-score} = \frac{2 \times precision \times recall}{precision \times recall} \times 100\% \tag{12}$$

$$\text{Accuracy} = \frac{TP + TN}{(TP + FN) + (FP + TN)} \times 100\% \tag{13}$$

We employed the holdout technique, as referenced in [29] and [38], to assess the performance of our banana breeds. To ensure an accurate evaluation, we used the criteria outlined in equations (10) - (13). This approach helped us effectively measure the model's accuracy and reliability.

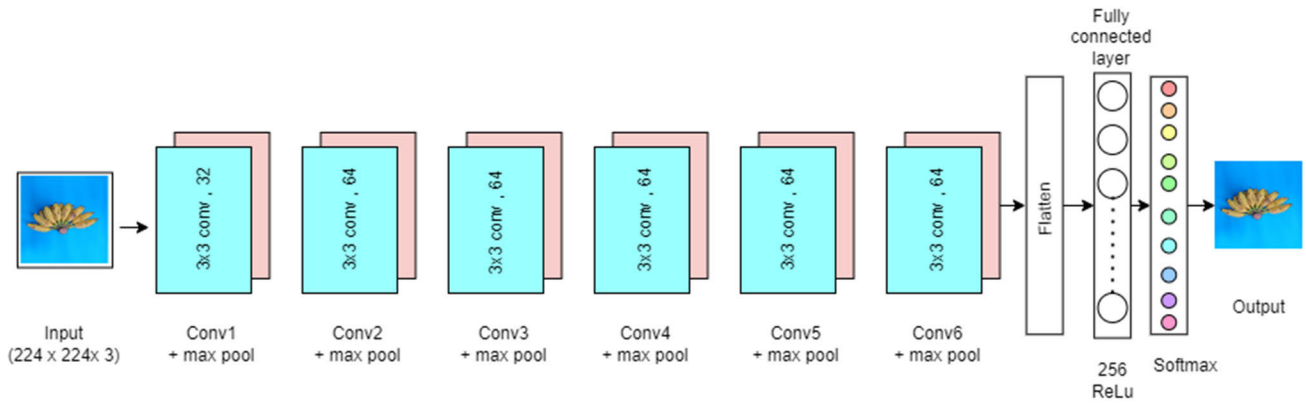


FIGURE 6. Customized CNN model architecture.

V. EXPERIMENTAL EVALUATION

Our approaches for banana breed recognition, as shown in Fig. 3, are applied to research projects. The initial stage of our approach is to take an image of a banana breed. The input image was resized to a size of 224 × 224 pixels. According to Fig. 3, this completes the preprocessing of the data. Every breed of banana has been augmented. Enhancing the training set's evaluation without adding extra photos is possible with the help of image augmentation. Following Fig. 3, this generates a processed data set. We used 5,100 images of six different banana breeds. We conducted an evaluation using the split holdout approach [29]. Three parts of the dataset were randomly selected: 10% for validation, 10% for testing, and 80% for training. We used the customized CNN model together with VGG16, ResNet50, Mobile-Net, and Inception-v3 as our seven CNN model algorithms. The deep learning models were specifically selected for classifying different banana breeds rather than being chosen randomly. The backend was developed using TensorFlow [39] and Keras [40], both high-level frameworks built on TensorFlow. Along with CNN, the design also contained convolutional, pooling, fully connected, and flattened layers [41]. The output's non-linearity was added using an activation function, which is essential to the CNN's efficacy [41]. This could refer to the ReLU activation function. Training processes were applied to each classification model, as shown in Table 4, to estimate the ranges for each key component. Confusion matrices were constructed for each model.

Table 5 displays the performance evaluation of five machine learning models: k-NN, SVM, Random Forest, Decision Tree, and an ensemble model combining SVM and Random Forest. All models were trained on deep features extracted via a CNN-based encoder rather than raw image pixels to ensure a fair comparison. Performance was assessed using accuracy, precision, recall, and F_1-score. K-NN achieves the highest accuracy at 0.84, making it the most dependable model for accurate predictions. SVM ranks second with an accuracy of 0.82, demonstrating strong performance, albeit slightly lower than k-NN. Random Forest presents a moderate accuracy of 0.78, indicating it operates less efficiently than k-NN and SVM, yet remains competently functional. However, despite these tuning efforts, the Decision Tree recorded the lowest accuracy (0.54) among all tested models. This underperformance is not caused by the dataset, as other algorithms achieved good accuracy.

TABLE 4. Specifications of the models used for each of the CNN models.

Algorithm	Specifications
VGG16	Total parameters: 21139014 Trainable parameters: 6424326 Non-trainable parameters: 14714688 Batch number = 32 Epoch count = 50
ResNet50	Total parameters: 49279622 Trainable parameters: 25691910 Non-trainable parameters: 23587712 Batch number = 32 Epoch count = 50
Mobile-Net	Total parameters: 16075718 Trainable parameters: 12846854 Non-trainable parameters: 3228864 Batch number = 32 Epoch count = 50
Inception-v3	Total parameters: 34911782 Trainable parameters: 13108998 Non-trainable parameters: 21802784 Batch number = 32 Epoch count = 50
MobileNetV3	Total params: 3002118 Trainable params: 5766 Non-trainable params: 2996352 Batch number = 32 Epoch count = 50
Hybrid CNN-ViT	Total params: 2509894 Trainable params: 251910 Non-trainable params: 2257984 Batch number = 32 Epoch count = 50
Customized CNN model	Total parameters: 541574 Trainable parameters: 541574 Non-trainable parameters: 0 Batch number = 32 Epoch count = 50

TABLE 5. Performance metrics evaluation for each of the machines Learning Model.

Model	Accuracy	Precision	Recall	F_1-Score
k-NN	0.84	0.84	0.84	0.84
SVM	0.82	0.82	0.82	0.82
Random Forest	0.78	0.78	0.78	0.78
Decision Tree	0.54	0.54	0.54	0.54
Ensemble methods (SVM + Random Forest)	0.87	0.87	0.87	0.87

TABLE 6. Performance metrics evaluation for each of the CNN models.

Algorithm	Accuracy (%)	Precision	Recall	F_1-Score
VGG16	97.71	0.98	0.98	0.98
ResNet50	71.88	0.74	0.72	0.72
Mobile-Net	98.12	0.98	0.98	0.98
Inception-v3	97.08	0.97	0.97	0.97
MobileNetV3	93.23	0.98	0.97	0.97
Hybrid CNN-ViT	88.00	0.88	0.88	0.88
Customized CNN model	99.37	0.99	0.99	0.99

Instead, it results from the structural limitations of Decision Trees, which restrict their ability to model complex feature interactions effectively. They rely on greedy, locally optimal splits that miss globally effective partitions and exhibit high variance, as small changes in training data drastically alter the tree structure and outcomes. Moreover, they have limited ability to represent complex feature interactions unless the tree grows excessively deep, which in turn increases overfitting. In contrast, k-NN adapts flexibly to local neighborhoods (achieving 0.84 accuracy) and SVM leverages global margin optimization with kernel functions (achieving 0.82 accuracy), both of which enable stronger generalization. This pattern is reflected in the different metrics, precision, recall, and F_1-score, where k-NN consistently outperforms the rest. The ensemble model, which integrates SVM and Random Forest, achieved an accuracy of 0.87 along with precision, recall, and F_1-score all at 0.87. This suggests that combining classical models can yield a noticeable performance improvement over individual models, thereby narrowing the gap with deep learning architectures. Overall, the ensemble method is the most balanced and effective model for this task.

Table 6 demonstrates the efficiency of the model analysis, revealing that transfer learning approaches outperformed others when applied to our dataset. Their accuracy range is 71.88% to 99.37%. Among the available options, selecting the optimal model is crucial. Table 7 highlights the experimental results for the best configuration, which was achieved through hyperparameter tuning.

Table 7 compares accuracy across different experiments to identify the best Customized CNN model. The results indicate that the highest accuracy was achieved in Experiment No. 4, where the CNN model was trained for 50 epochs using input images of 224 × 224 pixels. Experiment No. 4, the full CNN model consists of a series of 5 Convolutional (Conv) layers that have the following number of filters: 32, 64, 64, 64, 64. All of the Convolutional Layers used a 3 × 3 kernel, were activated with the ReLU activation function, and alternated with a 2 × 2 Max Pooling Layer. The model's architecture includes a dense layer with 256 units. Ten percent was set aside for testing, ten percent was put aside for validation, and eighty percent was used for training. On the test data, the model's accuracy was 99.37%. The customized CNN model's training history is displayed in Fig. 7.

Deep learning model gives better performance than classical machine learning models. Deep learning models outperform classical machine learning models because they can automatically learn complex features and patterns from large amounts of data. They use a number of layers (neural networks) to obtain a high-level representation, increasing the accuracy of image and speech recognition tasks. Classical machine learning models frequently underperform compared to deep learning models, as they have difficulty handling large, high-dimensional, and unstructured data such as images. They rely considerably on hand-extracted features, limiting their ability to learn complex patterns.

Table 6 evaluates seven CNN-based architectures: a customized CNN model, VGG16, ResNet50, Mobile-Net, Inception-v3, MobileNetV3, and a hybrid CNN-ViT model. Equations (2) - (5) were used to calculate the accuracy, precision, recall, and F_1-score used to evaluate the performance of each model. According to Table 6 statistics, the customized CNN model had the best accuracy at 99.37%. The customized CNN model outperformed the Mobile-Net and VGG16 models in banana breed recognition by 1.25% and 1.66%, respectively. The other algorithms also demonstrated performance, with VGG16, ResNet50, Mobile-Net, and Inception-v3 achieving accuracies of 97.71%, 71.88%, 98.12%, and 97.08%, respectively, indicating that these models are also effective in recognizing banana breeds, though less accurate than the customized CNN model. To ensure edge device readiness and efficiency, we tested MobileNetV3, which achieved an accuracy of 97.23%, with a precision of 0.98, recall of 0.97, and F_1-score of 0.97. We also implemented a hybrid CNN-ViT architecture to capture global texture variations in banana skins. This model yielded an accuracy of 88.00%, precision of 0.88, recall of 0.87, and F_1-score of 0.87. While its performance was lower than CNN-based models, it shows promise for tasks requiring attention to fine-grained, global patterns.

The customized CNN model architecture consists of three convolutional layers (each followed by ReLU activation and max pooling), a flattening layer, and two fully connected dense layers with dropout for regularization. The final output layer uses softmax activation for classification. This architecture was designed to strike a balance between depth and practical deployment, while achieving high accuracy. The demonstration of correctly and incorrectly recognized images is displayed in Fig. 8.

A customized CNN is beneficial for pre-trained models like VGG16, ResNet50, Mobile-Net, Inception-v3,

TABLE 7. Customized CNN model analysis.

Experiment No.	Epoch Count	Image Size	No. of Kernels (Conv)	Fully Connected Layer	Training Data	Validation Data	Test Data	Accuracy
1	50	224 × 224 pixels	32, 64, 64	256	80%	10%	10%	92.71%
2			32, 64, 64, 64		80%	10%	10%	98.10%
3			32, 64, 64, 64, 64		70%	10%	20%	98.12%
4			32, 64, 64, 64, 64		80%	10%	10%	99.37%
5			32, 64, 64, 64, 64, 64		80%	10%	10%	98.75%

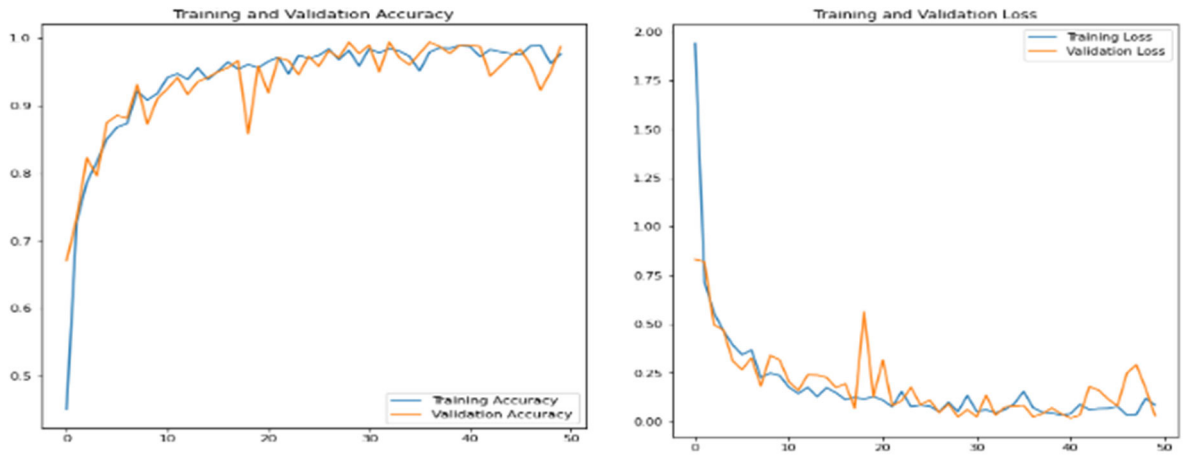


FIGURE 7. Training history of the customized CNN model.

MobileNetV3, or Hybrid CNN-ViT mainly when the dataset contains distinctive features that distinguish it from other picture datasets like ImageNet. When hardware resources are restricted, a customized CNN enables building more lightweight models with fewer parameters and layers, leading to shorter training and estimated times. This adaptive method makes a customized model extremely flexible to particular tasks and resource limitations by avoiding needless complexity, successfully capturing domain-specific features, and preventing overfitting on limited datasets. Pre-trained models such as VGG16, ResNet50, MobileNet, Inception-v3, MobileNetV3, and Hybrid CNN-ViT are developed for general image datasets like ImageNet, which makes them less effective for specific tasks. Customized CNNs tend to perform better because they are tailored to the target dataset, enabling them to capture task-specific features more accurately.

We expanded the banana picture dataset in the three-stage process from 5,100, 10,200 and 15,300 images, utilizing additional augmentation approaches in order to gain a better understanding of how dataset size affects model performance. The customized CNN was then retrained in both scenarios so that we could compare its robustness and accuracy. In both cases, the customized CNN was trained with the same kernel structure (32, 64, 64, 64, 64) and a split for training (80%), validation (10%), and test (10%) sets. The model reached its maximum accuracy of 99.37% with 5,100 images, 99.22% with 10,200 images and 99.14% with 15,300 images. This little variation shows that the model is already quite robust and tuned, sustaining good performance across a wide range of dataset sizes. The customized CNN consistently outperformed the baseline models in both scenarios, demonstrating its dependability for banana breed recognition. Fig. 9 presents a comparison of customized CNN accuracy across dataset sizes.

A. ENSUREMENT ROBUSTNESS

The robustness of an analytical method is its capacity to stay unaffected by small but deliberate changes in method variables. The results from k-fold cross-validation and the holdout method differ due to the way the data is used for training and testing. K-fold cross-validation provides a more reliable estimate by averaging the performance over multiple folds, which reduces the effect of any single data split. However, this comes at the cost of increased computation time, as the model is trained k times. On the other hand, the holdout method splits the dataset only once into training and testing sets, which makes it significantly faster and more efficient, especially for large datasets or complex models. While the results from holdout can vary depending on the split, enhanced holdout techniques like stratified sampling or repeated holdouts improve their stability. In practical scenarios where time, resources, or quick iteration are critical, holdout approaches prove better than k-fold cross-validation by offering a faster and simpler evaluation process without a significant loss in accuracy, particularly when working with large and representative datasets. We applied a stratified k-fold cross-validation to ensure that

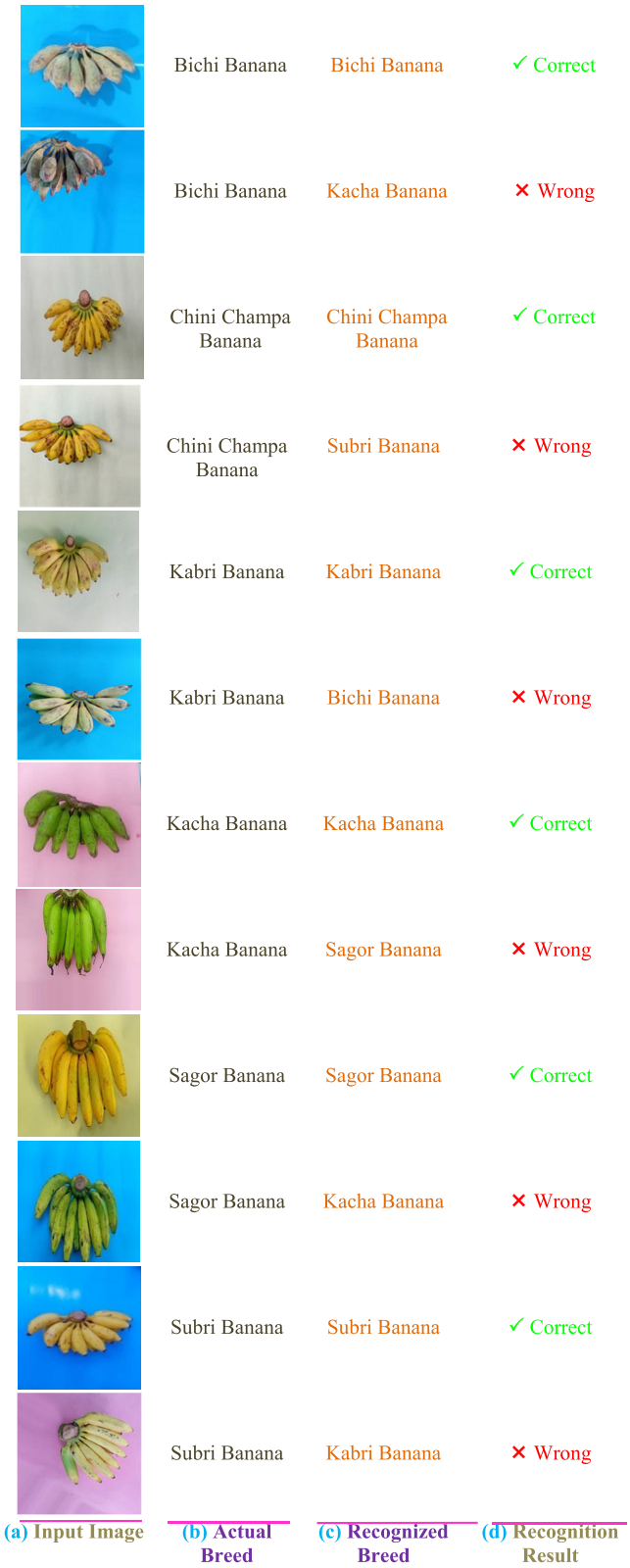


FIGURE 8. Demonstration of correct and incorrect recognition of banana breed images. (a) Input image, (b) Actual breed, (c) Recognized breed, (d) Recognition result.

classes were balanced, so images from the same bunch never entered the train and validation sets. For holdout, we reserve a stratified 10% independent test set before any training and only use it once during final reporting; no model selection is performed on it.

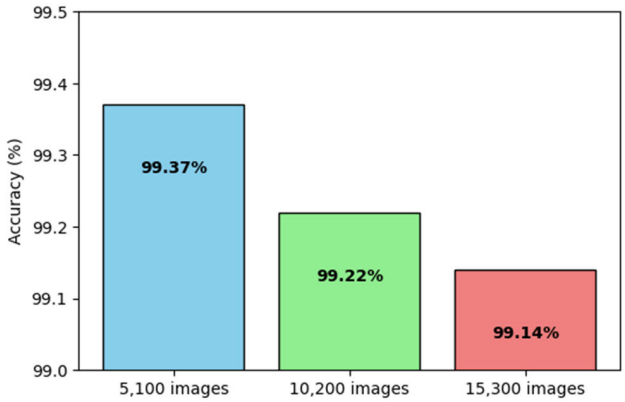


FIGURE 9. Comparison of customized CNN accuracy across dataset sizes.

Banana breed recognition presents unique challenges compared to general fruit or agricultural classification. Different banana breeds often exhibit high intra-class similarity in appearance, with only subtle differences in size, curvature, color tone, and surface texture. Furthermore, these traits are influenced by regional and environmental conditions, which adds to the complexity of distinguishing breeds. These characteristics justify the study of banana breed recognition as a distinct and valuable research problem rather than a subset of general fruit classification.

To ensure rigorous evaluation, we trained our best-performing model using both k-fold cross-validation and the holdout method. The customized CNN achieved 99.37% accuracy with the holdout method and 98.16% accuracy with k-fold cross-validation, a difference of only 1.21%. This consistency confirms the robustness and reliability of the proposed model across different validation strategies. By reporting both results, we ensure that our evaluation remains scientifically rigorous while also acknowledging computational efficiency considerations. Fig. 10 illustrates the comparison between k-fold cross-validation and the holdout method used to evaluate the accuracy of the customized CNN model.

We have provided confusion matrices for the ResNet50 (see Fig. 11 (a)) and the Customized CNN model (see Fig. 11 (b)) to better understand classification performance across various banana breeds. These matrices help identify which breeds are more prone to misclassification. For example, the Customized CNN model occasionally misclassifies Kabri_Banana as Chini_Champa_Banana, Subri_Banana, or Sagor_Banana. In contrast, the ResNet50 model demonstrates stronger overall accuracy and fewer misclassifications across all classes.

VI. COMPARATIVE ANALYSIS OF RESULTS

Our suggested method's efficacy in recognizing banana breeds needs to be evaluated by comparing it with current

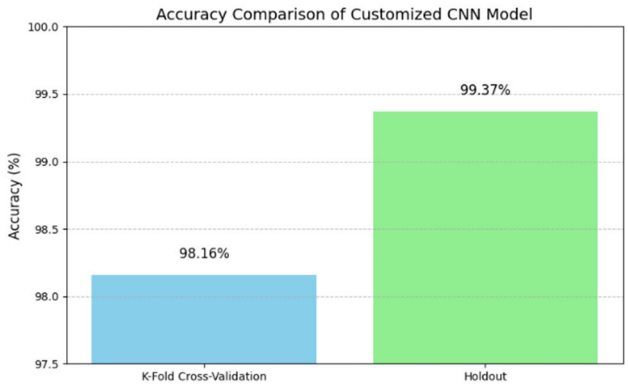
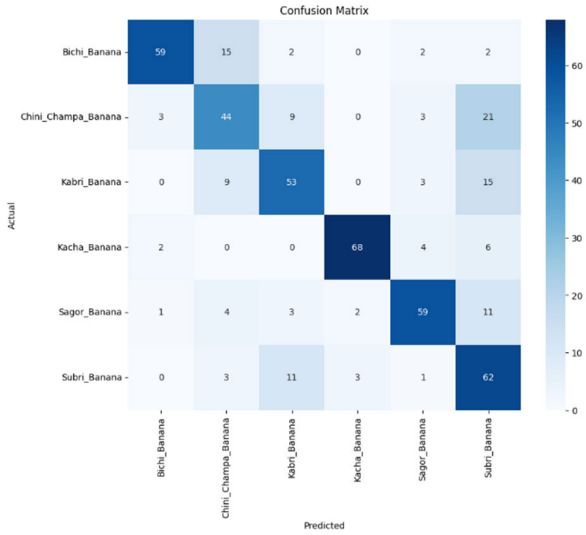


FIGURE 10. K-fold cross-validation and holdout method used to evaluate the accuracy of the Customized CNN model.

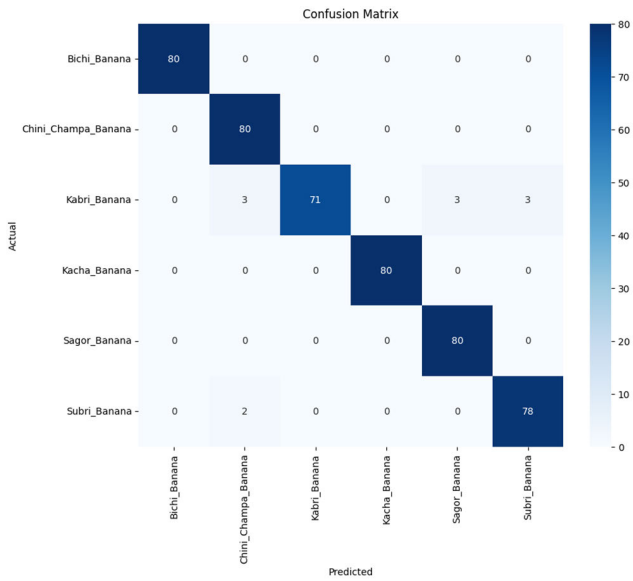
research. It is impossible to compare our technique's features because there isn't enough research on banana breed recognition. In this respect, state-of-the-art works, among many others, use deep learning for banana quality identification, coffee bean defect identification, grape variety recognition, and potato leaf disease identification. Through the review of the literature, most of the published works presented just an abstract of machine vision and deep learning concepts, either for fruit species or disease identification, lacking enough numerical data or comparisons with relevant research. The evaluation of the results of different efforts is further complicated by the absence of a common database containing samples from different fruit types and diseases. However, we attempted to compare our work to others by considering dataset size, approach, accuracy, and other factors. Table 8 compares and contrasts our study with previous research.

Rahman et al. [12] applied deep learning techniques to potato breed recognition, using 6,000 images spanning 10 classes. The customized CNN model, which had an accuracy of 94.84%, was the best. Tiwari et al. [13] concentrated on detecting potato leaf disease using logistic regression and VGG19, with 97.8% accuracy on a dataset of 2,152 photos in 3 classifications. Huang et al. [15] used a CNN algorithm to achieve 93% accuracy on a dataset of 72,000 photos divided into two classes. Rimi et al. [42] used Inception-v3, achieving 98.6% accuracy on a dataset of 5,400 images across nine classes. Al Haque et al. [43] conducted research focused on identifying banana strain and quality using a dataset comprising 10,000 images, categorized into five distinct classes. This study was conducted in the field of computer vision, and the CNN classifier has the best accuracy at 99.37%.

As was already indicated in Section II, no research has been done specifically on banana breeds. Due to the lack of studies on banana breed recognition, it isn't easy to accurately compare the advantages of our method. Instead, except for providing sufficient numerical results and comparing them to those of other comparable works, the majority of the pertinent works included above are limited to illustrating concepts of machine vision strategy for fruit disease, breed, and quality



(a) ResNet50 Confusion Matrix



(b) Customized CNN model confusion Matrix

FIGURE 11. Confusion Martix: (a) ResNet50 Confusion Matrix and (b) Customized CNN model confusion Matrix.

recognition. Research on fruit disease, breeds, and quality recognition has seen some encouraging developments in recent years, but thorough evaluations of performance based on real-world signals are still lacking. To assess the relative qualities of our study, we have analyzed the numerical results concerning fruit diseases, breeds, and quality recognition in Table 8. We may assert that our achieved accuracy of over 99% is good and promising enough for the case shown in Table 6.

VII. SYSTEM IMPLEMENTATION

We build simple websites based on our high-performance model. The website was built using Python for the backend and React.js for the front-end, which is a basic yet strong

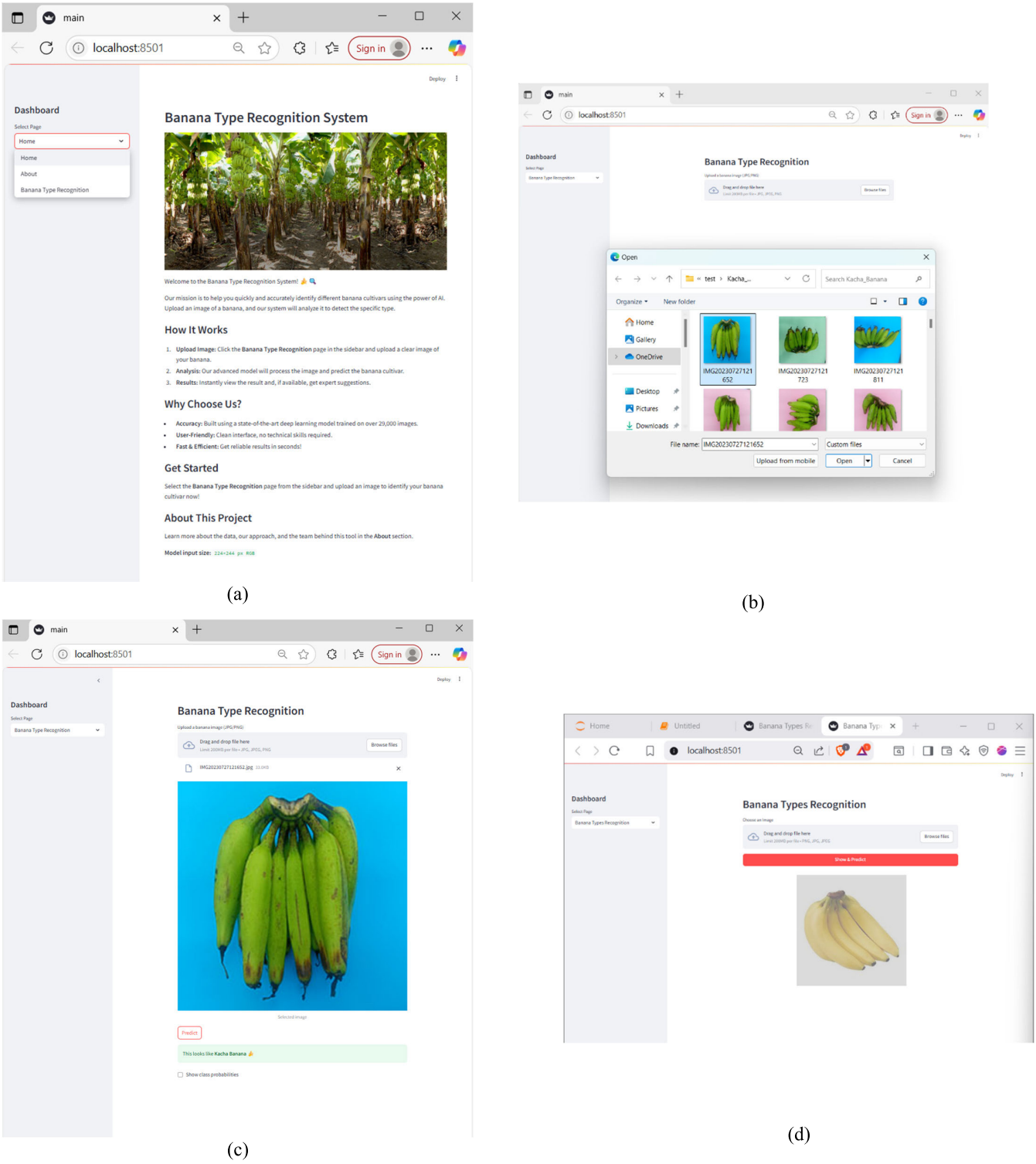


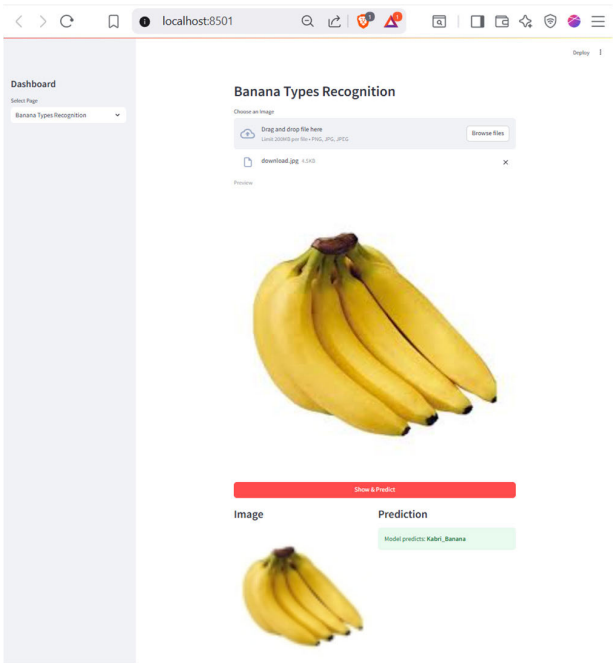
FIGURE 12. Simple website for banana breed recognition. (a) Website home (b) Select banana breed (c) Breed recognition. (d) Select banana breed from inline banana image (e) Online banana image breed recognition.

technology stack. React.js is a prominent front-end package that has steadily evolved into the JavaScript community's preferred framework for modern web development. We installed this webpage on our local laptop.

This is seen in Fig. 12. The primary landing page of the website, where users begin their trip, is shown in (a). (b) demonstrates that the users can choose from a number of banana breeds. Users can drag and drop unknown banana

TABLE 8. Results of comparison of our work with other notable works.

Method/work done	Fruit	Problem Dealt with	Size of image Data Set	No. of Classes	Best Performing Model	Accuracy
Proposed Work	Banana	Breed recognition	5,100	6	Customized CNN	99.37%
Rahman et al. [12]	Breed recognition	Breed recognition	6,000	10	Customized CNN	94.84%
Huang et al. [15]	Coffee bean	Quality recognition	2000	2	CNN	93%
Rimi et al. [42]	Legume	Legume species recognition	5,400	9	Inception-v3	98.6%
Al Haque et al. [43]	Banana	Strain and quality recognition	10,000	5	CNN	93.4%



(e)

FIGURE 12. ***(Continued.)*** **Simple website for banana breed recognition. (a) Website home (b) Select banana breed (c) Breed recognition. (d) Select banana breed from inline banana image (e) Online banana image breed recognition.**

breeds or submit images of bananas. (c) show the technique used by the website's technology to recognize the banana breed using the top-performing model. (d) illustrates the option to select a banana breed directly from an online image of a banana. (e) presents the online banana image breed recognition process. We tested our system with banana images from our dataset and online and found that our system can accurately recognize banana breeds. It is worth noting that if the reader is interested, he/she can request the source code from the respective author and run it on his/her computer.

VIII. CONCLUSION AND FUTURE WORK

This study demonstrates the effectiveness of deep learning models in recognizing different breeds of bananas. Deep learning techniques demonstrate significant potential for accurately recognizing different banana breeds by learning complex patterns from large datasets. This study explicitly addresses banana breed recognition, which is particularly difficult given the high degree of similarity among banana varieties and their susceptibility to regional and environmental conditions. We also provide one of the first collected and annotated benchmark datasets for banana breed recognition that we are aware of, laying the groundwork for subsequent research and reproducibility. In addition, our research offers important insights into which models are most suited for this particular task by contrasting five machine learning model and seven deep learning architectures and providing a high-performing customized CNN. The deep learning models VGG16, ResNet50, Mobile-Net, Inception-v3, MobileNetV3, Hybrid CNN-ViT, and a customized CNN have effectively distinguished various banana breeds based on their unique visual attributes. Among these, the customized CNN model achieved the highest accuracy at 99.37%, showing its robust performance. We developed a web application that enables users to identify banana breeds, allowing them to recognize the breed themselves both through the web application and by checking in person. This research not only advances the application of deep learning in agricultural technology but also provides a foundation for further improvements and practical implementations. In the future, we will work on obtaining larger datasets and working on banana quality, ripeness, and disease detection. Future work could include developing a mobile app to allow farmers to upload banana images for automatic breed identification, simplifying the process and increasing efficiency.

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